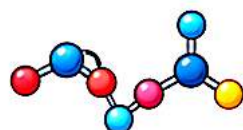


SIGNALING TRANSITION KINETICS: COMBO TREATMENT (BRAFi+MEKi) IN MELANOMA

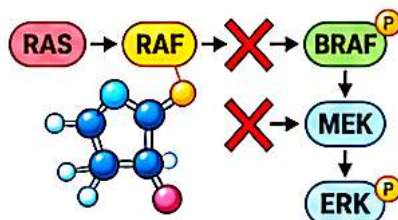


COMBINED BRAFi + MEKi TREATMENT (e.g., Dabrafenib + Trametinib)

● PHASE 1: ACUTE SUPPRESSION (6h - 24h)



- Immediate Therapeutic Efficacy: Confirms Target Blockade.

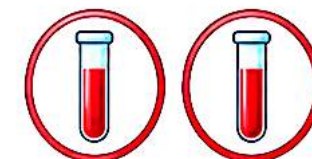
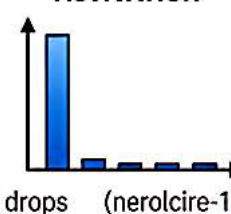


MAPK PATHWAY SHUTDOWN

KSEA SCORES
(Target Phosphosites)

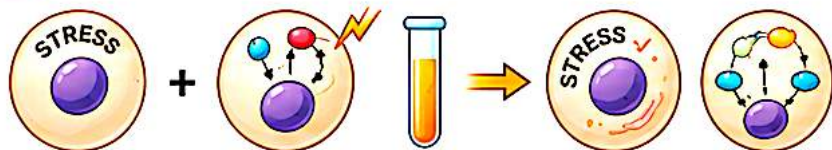


ERK1/2
ACTIVATION



CELLS AIM IN FULL
SUPPRESSION

● PHASE 2: ADAPTIVE REWIRING (48h - 72h)



- Emergency Adaptation: Cells Bypass the Block.

FEEDBACK
LOOP LOSS

EMERGENCY
ACTIVATION
OF PARALLEL
ROUTES

PI3K/AKT/mTOR
SCORES



RTK EXPRESSION
(AXL, PDGFR)

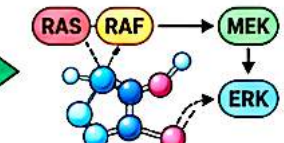


REBOUNING KINASE ACTIVITIES (e.g.)

● PHASE 3: CONSOLIDATED RESISTANCE (3d - 90d)



- Stable Resistance: Permanent Evolutionary Shift.



FULLY INTEGRATED,
ALTERNATIVE SIGNALING

MULTI-OMICS
HEAT MAP



PHOSPHOPROTEOMIC
REBOUND

Long-term RNA-seq
GENE EXPRESSION CHANGES
(GSE110054)

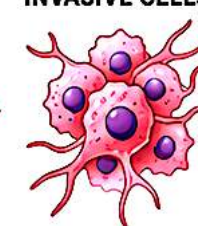


Gene expression signatures:
stable, concordant signatures^(1,2)

PHENOTYPIC
SWITCHING



INVASIVE CELLS



6h

24h

48h

72h

3d

11d

21d

90d